

Interfaces of FFGFT to Other Frameworks

Comparison and bridge documents, points of contact and overlapping quantities

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Interfaces of FFGFT to Other Frameworks

Abstract. The corpus contains comparison and bridge documents for eleven frameworks, spread over more than twenty documents. This document collects them: for each framework the document numbers, the mathematical point of contact, the status and the open item. The quantities where several frameworks actually overlap follow, each with its route, as a template for a comparison table. Nothing new is computed; all numbers come from the documents named and are recomputed in the check script.

List of symbols

- [B]** algebraically proven
- [K]** numerically confirmed
- [S]** structurally plausible, derivation open
- [X]** checked and negative
- [Q]** taken from an external source, cited

1 Principle

FFGFT derives from ξ . Measured values and Standard Model quantities serve for conversion and testability, not for derivation; the cosmic sector stays bracketed (P39). Quantities of external frameworks appear in all documents named below as cited inputs **[Q]** and do not stand on equal footing with our own posits. A point of contact counts as a bridge only when the same structural object follows from each framework's own geometry and an explicit map between the derivations exists; the criterion is from Doc. 283.

2 Overview

Framework	Documents	Point of contact	Status
HLV (Krüger)	271, 272, 276, 282, 283, 285, 294, 297	icosahedral cut-and-project quasicrystal versus $T^4/\mathbb{Z}_3$; dimension bridge $6 = \mathbf{3} \oplus \mathbf{3}'$ in A_5 ; CP divisibility	bridge [K] , fork at the symmetry order
OPH (Mueller)	364, 374, 375, 376	spectrum of the icosahedral carrier; energy-length bridge $E_{\text{bit}} = \hbar c/L$; v/E_p ; coefficient 8π	bridge equations [B] , scales open on the other side [Q]
Hyperbit, $G(6, \mathbb{Z}_3\mathbb{C})$ (Matzke)	324, 326, 336	$\mathbb{Z}_3$ structure and $\text{GF}(9)$; bit energy and the Landauer threshold	algebraic bridge [B] , bit energy system-dependent [K]
XYLATIC (Phillips)	252	number-theoretic structures of the σ function	structural comparison
RA/PMT (Guevara)	269	three-layer scheme, perception level	structural comparison
Vopson	251	computational-universe hypothesis, second law of infodynamics	different primitives
Matsas et al.	105	the number of fundamental constants	classification
ΛCDM	309	scale anchor: both frameworks need exactly one number for the SI coupling	shared situation [K]
Kagome lattice	361	four structural parallels; mass as geometric freezing	analogy [S]
SM Lagrangian	049	number of fields and free parameters against one equation	comparison of form
Literature in general	345	classification against published approaches	classification

UIFT (Teker) is touched in Docs. 013 and 257 through the bit mass, but has no comparison document of its own yet.

3 Quantities where frameworks overlap

This table is the template for a comparison table. Every value comes with its route; without it the value is not comparable (Doc. 375).

Quantity	FFGFT value	Dev.	Route
α^{-1}	$3700/27 = 137.037$	+7.6 ppm	geometry from $T^4/\mathbb{Z}_3$ (Doc. 007), Galois confirmation (Doc. 338)
$\sin^2 \theta_W$ (on-shell)	$2/9 = 0.22222$	-0.52 %	A_5 structure (Docs. 293, 372)
λ_{CKM}	$\xi^{1/6} = 0.22602$	+0.46 %	Galois bridge (Doc. 336)
v/E_p	2.0389×10^{-17}	+1.10 %	v from ξ (R104), E_p from our own chain (Doc. 375)
$E_{\text{bit}}(L)$	$\hbar c/L$	—	system-dependent, no universal bit value (Docs. 257, 326)
Dimension	$3+1$ from $T^4/\mathbb{Z}_3$	—	posit; $6 = \mathbf{3} \oplus \mathbf{3}'$ as the bridge to HLV (Doc. 285)
Cosmic sector	—	—	bracketed (P39, Doc. 309)

4 What follows from this

- Two frameworks are connected through more than one quantity: HLV through the A_5 structure and the dimension, OPH through the carrier, the scale and v/E_p . The rest are single comparisons.
- The overlaps lie almost entirely in the lepton sector and in the geometry. The hadron sector appears in no comparison; that is not a selection but the state of the other sides.
- α^{-1} and v/E_p are currently the only quantities for which more than one framework states a number. They are therefore the only rows of a comparison table that data can tell apart.
- The scale anchor is the same situation in a different place for every framework (Doc. 309); it is not a distinguishing feature.

Tool and method

Claude (Anthropic) was used as a tool for: assembling the document list, writing and running the check script, drafting the $\text{\LaTeX}$ sources, and maintaining the changelog. The tool computes and writes; the author decides what is computed and written, judges every result and determines what it means physically. The check script verifies that every cited document is present in the corpus and recomputes every number of the second table from FFGFT quantities.

5 Result

1. Eleven frameworks with comparison or bridge documents, two of them with several points of contact **[K]**.
2. Seven quantities where frameworks overlap, each with route and deviation **[K]**.
3. Only α^{-1} and v/E_p are given a number by more than one framework.
4. The hadron sector appears in no comparison.

Check script: 2/python/Dok377_Skripte/pruef_377_schnittstellen.py (20/20)

Register (Doc. 190): no entry.

Bibliography

- [1] Docs. 271, 272, 276, 282, 283, 285, 294, 297 (HLV).
- [2] Docs. 364, 374, 375, 376 (OPH).
- [3] Docs. 324, 326, 336 (Hyperbit, $\mathbb{Z}_3\mathbb{C}$).
- [4] Docs. 049, 105, 251, 252, 269, 309, 345, 361.
- [5] Docs. 007, 013, 257, 293, 338, 372 (FFGFT sources of the numbers).